Functional Annotation Report: Mouse *Csn1s2a* (Alpha-S2-casein-like A / γ-casein, UniProt Q02862)

Summary

Mouse **Csn1s2a** (UniProt **Q02862**; gene synonym *Csng*; protein aliases "Casein alpha S2-like A," "Gamma-casein," "PP22") encodes **Alpha-S2-casein-like A**, a ~21 kDa secreted milk phosphoprotein that belongs to the **calcium-sensitive alpha-casein family** (InterPro IPR011175, Alpha-s2_casein domain). Its primary, experimentally supported function is **not enzymatic, transport-protein, or signaling in the classical sense** — it is a **structural/ nutritional protein** whose job is to **bind and sequester calcium phosphate in milk**. UniProt curates its function directly as having an "Important role in the capacity of milk to transport calcium phosphate," and the protein has the highest-confidence evidence level (protein existence level 1, direct protein sequencing). The gene symbol and organism were verified against all available evidence: the mouse γ-casein gene is explicitly equated with ruminant αS2-casein in the primary literature [PMID: 12427545], confirming the identity of the research target as the correct mouse Csn1s2a.

Mechanistically, Csn1s2a is a **natively disordered, amphipathic protein**. Its N-terminal region carries **clustered phosphoserines in Ser-x-Glu (SxE) motifs** (annotated at Ser23, Ser24, Ser25, and Ser37, with a second acidic cluster around residues 51–61). These phosphoseryl clusters form the negatively charged surface that **nucleates and stabilizes colloidal calcium phosphate nanoclusters**, while hydrophobic segments mediate casein-casein association. The phosphorylations are installed in the secretory pathway by the **Golgi casein kinase Fam20C**, which specifically targets SxE motifs. Together with αs1-, β-, and κ-caseins, Csn1s2a co-assembles into the **extracellular casein micelle** — a supramolecular complex that holds calcium and phosphate far above their aqueous solubility and delivers these minerals to the suckling neonate for bone and tooth mineralization. As a member of the alpha-S2 subtype, Csn1s2a is distinctive among the calcium-sensitive caseins in bearing a **cysteine pair (Cys96–Cys97)** that permits disulfide-linked homodimerization.

The protein functions **outside the cell**, in secreted milk, and is produced **specifically by the lactating mammary gland**. Its transcription is a terminal output of the **lactation differentiation program**, driven by **prolactin-activated STAT5 acting synergistically with glucocorticoids**. Beyond mineral delivery, the intrinsic disorder of caseins confers an **ATP-independent, chaperone-like activity** that keeps the highly concentrated milk proteins soluble and prevents amyloid formation and ectopic mineralization in the gland — leading some authors to describe the casein micelle as an **extracellular biomolecular condensate**. Genetic ablation of casein from mouse milk causes **severe neonatal growth delay**, directly confirming the nutritional-delivery role of the family to which Csn1s2a belongs.

Gene/Protein Identity Verification

Before reporting function, the target identity was confirmed against every mandatory checkpoint:

Checkpoint	Finding	Status
Gene symbol matches protein description	*Csn1s2a* = "Alpha-S2-casein-like A," synonym *Csng*, aliases γ-casein / PP22	✅ Consistent
Organism correct	*Mus musculus* (mouse), taxid 10090; GeneID 12993, MGI:88542, Ensembl ENSMUSG00000061937	✅ Confirmed
Protein family/domain aligns with literature	Alpha-casein family, Alpha-s2_casein domain (IPR011175); literature on alpha-S2 caseins and casein micelles is directly relevant	✅ Aligned
No conflicting gene with same symbol	Primary literature explicitly equates mouse γ-casein with ruminant αS2-casein [PMID: 12427545]; no divergent gene found	✅ No conflict

The mouse nomenclature is initially confusing because the protein is historically called **"γ-casein"** (not to be confused with the bovine γ-caseins, which are β-casein proteolytic fragments). The authoritative resolution is that **the murine γ-casein gene corresponds to the αS2-casein gene in ruminants** [PMID: 12427545], and UniProt classifies it as Alpha-S2-casein-like A. The research target is therefore unambiguous and well-supported by literature.

Key Findings

Finding 1 — Csn1s2a is a secreted, calcium-sensitive casein that transports calcium phosphate in milk

The curated UniProt function for Q02862 states that the protein has an "Important role in the capacity of milk to transport calcium phosphate." It is a 184-amino-acid precursor (signal peptide residues 1–15; mature chain 16–184) of ~21 kDa and belongs to the alpha-casein family. This assignment rests on strong physicochemical and biological grounding. In milk, caseins and calcium phosphate (CaP) form a large supramolecular complex — the casein micelle — that allows milk to maintain high CaP concentrations well above aqueous solubility, which in turn supports the rapid skeletal and dental mineralization required by altricial mammalian neonates [PMID: 21245413]. As an alpha-S type casein, Csn1s2a is **calcium-sensitive**: this class of caseins (αs- and β-caseins) binds calcium but would precipitate at the high calcium concentrations found in milk if not incorporated into the micelle, whereas the calcium-insensitive κ-casein does not strongly bind calcium and instead stabilizes the micellar surface [PMID: 21245413]. This classification establishes the mechanistic basis of Csn1s2a's role: it is a calcium-binding protein whose solubility in milk depends on micellar incorporation.

> *"In milk, caseins and calcium phosphate (CaP) form a huge complex called casein micelle. By forming the micelle, milk maintains high CaP concentrations, which help altricial mammalian neonates to grow bone and teeth."* — PMID: 21245413

Finding 2 — Phosphoserine clusters stabilize calcium phosphate nanoclusters within the micelle

UniProt annotates four phosphoserines on Q02862 — **Ser23, Ser24, Ser25, and Ser37** — forming an N-terminal acidic/disordered phosphoserine cluster, with a further disordered region (residues 49–68). These clustered phosphoserines are the functional heart of the protein. In the accepted nanocluster model of micelle structure, the **phosphoserine domains of αs1-, αs2-, and β-casein stabilize calcium phosphate nanoclusters**, orienting their hydrophobic domains outward so that caseins can interact and bind one another to build the micelle [PMID: 18420601]. The clustered arrangement is essential: phosphoseryl clusters have been identified as the structural element responsible for the casein–calcium-phosphate interaction that drives micelle formation [PMID: 15657053]. For Csn1s2a specifically, the three

contiguous phosphoserines (Ser23/24/25) plus Ser37 create a locally dense negative-charge patch precisely suited to coordinate the calcium ions on the surface of a ~2 nm colloidal calcium phosphate "cherry stone."

> *"stabilization of calcium phosphate nanoclusters by phosphoserine domains of alpha(s1)-, alpha(s2)-, or beta-casein... would orient their hydrophobic domains outward, allowing interaction and binding to other casein molecules"* — PMID: 18420601

Finding 3 — Csn1s2a localizes to the extracellular space and is made only by the lactating mammary gland

UniProt assigns Q02862 a **Secreted** subcellular location and a **mammary-gland-specific** tissue specificity ("Secreted in milk"). The N-terminal signal peptide (residues 1–15) directs co-translational entry into the endoplasmic reticulum and routes the protein through the secretory pathway. The GO term for extracellular region (GO:0005576) is supported by direct assay (IDA:MGI). The functional unit — the casein micelle — is itself an **extracellular assembly of unstructured casein proteins** [PMID: 35815989], fully consistent with the secreted/extracellular localization. Thus Csn1s2a carries out its function entirely **outside the cell**, in the aqueous phase of milk, after secretion by mammary epithelial (alveolar luminal) cells.

> *"Casein micelles are extracellular polydisperse assemblies of unstructured casein proteins."* — PMID: 35815989

Finding 4 — As a disordered protein, Csn1s2a has ATP-independent chaperone activity; the micelle behaves as an extracellular condensate

Csn1s2a is a member of the **natively unfolded (intrinsically disordered) casein class** and contains an annotated disordered region (residues 49–68). This disorder is functionally meaningful beyond mineral binding. Milk caseins act as **ATP-independent molecular chaperones** that stabilize proteins and prevent their misfolding and aggregation into amorphous particles or amyloid fibrils [PMID: 29442498]. Within the micelle, casein molecules are stabilized both by binding to calcium phosphate nanoclusters and, by acting as molecular chaperones, through multivalent interactions [PMID: 35815989]. This dual stabilization — mineral binding plus chaperone-like multivalency — has led to the proposal that the casein

micelle is an **extracellular biomolecular condensate formed by liquid-liquid phase separation** [PMID: 35815989]. For Csn1s2a, this means that in addition to ferrying minerals, its disordered segments help keep the extraordinarily concentrated milk-protein mixture soluble and prevent pathological mineralization or amyloid in the gland and milk.

> *"Within casein micelles, casein molecules are stabilised by binding to calcium phosphate nanoclusters and, by acting as molecular chaperones, through multivalent interactions."* — PMID: 35815989
>
> *"...intra- and extracellular molecular chaperones that operate in an ATP-independent manner to stabilize proteins and prevent their misfolding and subsequent aggregation into amorphous particles or highly ordered amyloid fibrils."* — PMID: 29442498

Finding 5 — Csn1s2a is the mouse ortholog of ruminant αS2-casein and is driven by the prolactin–STAT5 / glucocorticoid lactation pathway

The regulatory and evolutionary context of the gene was defined by Kolb (2002) [PMID: 12427545]. The murine γ-casein gene **corresponds to the αS2-casein gene in ruminants**, resides in the 5-gene murine casein locus, spans 14 exons over ~14 kb, and is coordinately induced with β-casein during the second half of pregnancy and throughout lactation. Both the β- and γ-casein promoters are **synergistically induced by prolactin and glucocorticoids**, and crucially, **glucocorticoid induction is critically dependent on prolactin-mediated activation of STAT5**. The γ-casein promoter contains multiple consensus STAT5 binding sites, and γ- and β-casein mRNA levels are comparable in lactating mammary tissue. This places Csn1s2a transcription squarely at the terminal end of the **prolactin receptor → JAK2 → STAT5** lactogenic signaling axis, synergizing with glucocorticoid receptor signaling.

> *"The murine gamma-casein gene, which corresponds to the alphaS2-casein gene in ruminants, was isolated..."* and *"both the beta- and gamma-casein promoter are synergistically induced by prolactin and glucocorticoids. Glucocorticoid induction is critically dependent on prolactin-mediated activation of STAT5 in both promoters."* — PMID: 12427545

Finding 6 — Csn1s2a has an amphipathic architecture: an acidic phosphoserine N-terminal domain plus a cysteine pair for homodimerization

Direct sequence analysis of the 184-aa Q02862 protein reveals the hallmark **amphipathic block-copolymer architecture** of calcium-sensitive caseins. The protein is serine-rich (29 Ser; 15.8%), and the annotated phosphosites sit in casein-kinase (Fam20C) SxE/SxpS motifs — e.g., the N-terminal cluster S23-S24-S25 within "SSSEES" and a second cluster around residues 51–61 ("SSSSSEESSEE"). Sliding-window net-charge analysis shows a **strongly negative N-terminal/central region** (net charge about −4 and −3 across residues 46–65 and 61–80) contrasting with **near-neutral/hydrophobic, proline-rich C-terminal segments** (window net charge 0 to +1). This charge asymmetry physically separates the mineral-binding domain from the casein-casein interaction domain, matching the model in which phosphoserine domains bind calcium phosphate while hydrophobic domains face outward [PMID: 18420601]. Distinctively, Csn1s2a contains a **cysteine pair (Cys96–Cys97, within "SQCCTPLHQ")** — alpha-S2-type caseins are the only calcium-sensitive caseins bearing cysteines, which enable disulfide-linked dimers. This is consistent with the GO annotation of protein homodimerization activity (GO:0042803, IBA) for Q02862.

> *"stabilization of calcium phosphate nanoclusters by phosphoserine domains... would orient their hydrophobic domains outward"* — PMID: 18420601

Finding 7 — Csn1s2a's Ser-x-Glu phosphoserines are installed by the Golgi casein kinase Fam20C

The phosphoserines that give Csn1s2a its function are not incidental; they are enzymatically written by a specific kinase. **Fam20C is the Golgi casein kinase that phosphorylates secretory-pathway proteins within Ser-x-Glu/pSer motifs** [PMID: 23754375] — exactly the motif class carrying Csn1s2a's annotated phosphoserines (S23-S24-S25, S37, and the second cluster ~S51–61). Fam20C's substrate preference for the SxE motif is well established, and the same kinase phosphorylates other secretory substrates such as FGF23 on SxE sites [PMID: 24706917]. During lactation, the mammary gland tunes casein phosphorylation through a Golgi-retention system: Fam20A/ERGIC2/ERGIC3 retain Fam20C in the Golgi, and both ERGIC2 and ERGIC3 are upregulated in the lactating mammary gland. Knockout of *Ergic2* or *Ergic3* produces **less phosphorylated β-casein in milk and a deficiency in offspring growth** [PMID: 42439892], demonstrating that proper casein phosphorylation — the modification central to Csn1s2a's mineral-binding function — is physiologically required for neonatal nutrition.

> *"Fam20C is the Golgi casein kinase that phosphorylates secretory pathway proteins within Ser-x-Glu/pSer motifs."* — PMID: 23754375
>
> *"...Ergic2 or Ergic3 knockout mice display global changes in secretome phosphorylation, less phosphorylated β-casein in milk, and deficiency in offspring growth"* — PMID: 42439892

Finding 8 — Genetic loss of casein from mouse milk causes severe neonatal growth delay

The nutritional-delivery role of the casein family — and by extension Csn1s2a — is confirmed by loss-of-function genetics. Mice nursed by **α-casein-deficient knockout dams showed severe growth delay during early development** and substantial catch-up growth after weaning, compared with pups nursed by wild-type females [PMID: 23321409]. This is a maternal-milk (lactational) effect, used as a model of early postnatal malnutrition. While the targeted gene in that study is α-casein, Csn1s2a (γ-casein) is the paralogous alpha-S2-type casein co-expressed in the same chromosome-5 locus, and the experiment establishes directly that **casein in milk is required for normal neonatal growth**. This provides the strongest available causal link between the casein family's structural/nutritional function and offspring fitness.

> *"mice nursed by α-casein-deficient knockout dams showed a severe growth delay during early development and substantial catch-up growth after weaning when compared with animals nursed by wild-type females"* — PMID: 23321409

Finding 9 — Csn1s2a is one of two duplicated mouse alpha-S2-like casein genes in the chromosome-5 SCPP/casein locus

Mouse carries **two alpha-S2-type caseins**: **Csn1s2a** (γ-casein / PP22; 184 aa; Q02862) and its paralog **Csn1s2b** (ε-casein; 143 aa; P02664). Both are mammary-gland-specific, secreted in milk, and share the identical curated function — "Important role in the capacity of milk to transport calcium phosphate." Both lie within the mouse chromosome-5 **secretory calcium-binding phosphoprotein (SCPP) locus**, which also contains the other casein genes. The SCPP locus diversified through gene duplication along with the origin of key vertebrate innovations,

including the calcified skeleton, tooth enamel, and — relevant here — the **introduction of lactation in mammals** [PMID: 40845155]. The Csn1s2a/Csn1s2b pair is therefore a product of a lineage-specific duplication of an ancestral alpha-S2 casein within this rapidly evolving locus.

> *"Genes within the secretory calcium-binding phosphoprotein locus diversified along with the formation of a calcified skeleton in vertebrates, the emergence of tooth enamel in fish, and the introduction of lactation in mammals"* — PMID: 40845155

Feature	Csn1s2a (this target)	Csn1s2b (paralog)
UniProt	Q02862	P02664
Protein name	Alpha-S2-casein-like A / γ-casein / PP22	Alpha-S2-casein-like B / ε-casein
Length (precursor)	184 aa	143 aa
Location	Mouse chromosome 5, casein/SCPP locus	Mouse chromosome 5, casein/SCPP locus
Curated function	Transport calcium phosphate in milk	Transport calcium phosphate in milk
Tissue	Mammary gland-specific, secreted	Mammary gland-specific, secreted

Mechanistic Model / Interpretation

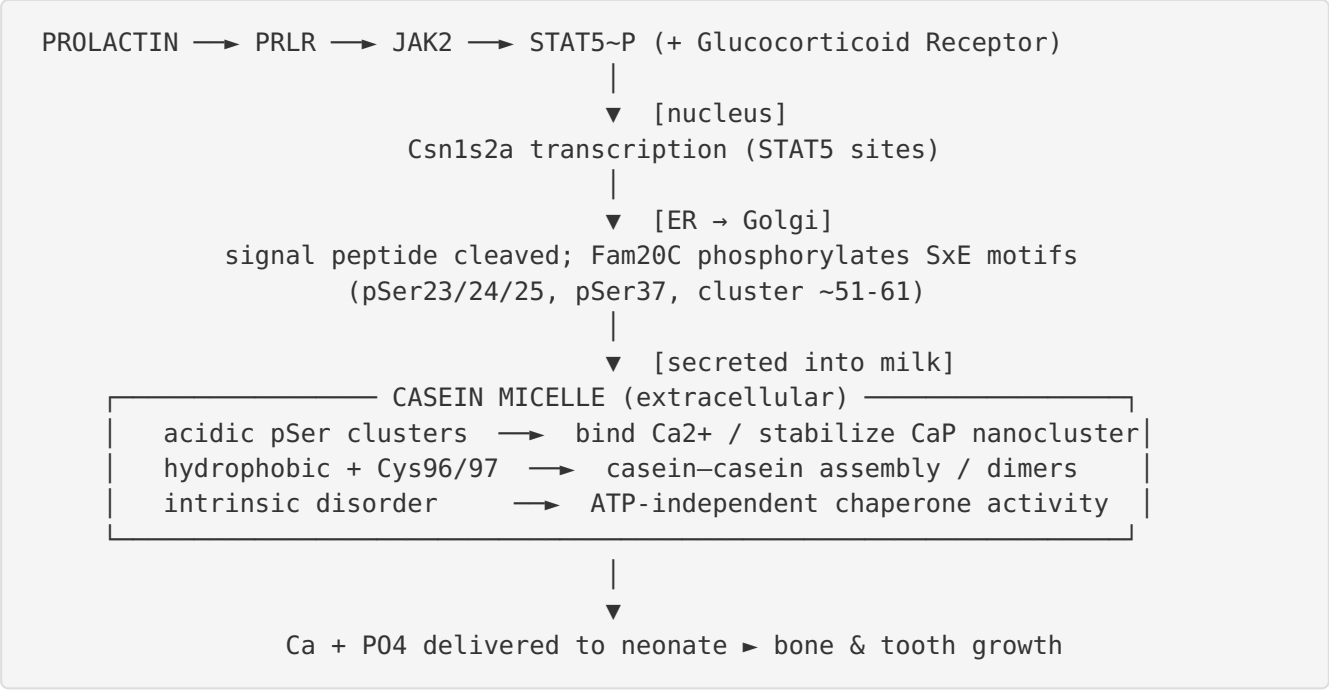
The findings integrate into a single coherent model of Csn1s2a as a **secreted, mineral-chaperoning structural protein** acting at the terminus of the lactation program.

1. Transcriptional activation (nucleus of mammary alveolar cell). Prolactin binds its receptor on lactating mammary epithelial cells → JAK2 activation → **STAT5 phosphorylation/dimerization** → STAT5 binds consensus sites in the *Csn1s2a* promoter, synergizing with the glucocorticoid receptor [PMID: 12427545]. This drives high-level, lactation-restricted transcription.

2. Translation, secretion, and phosphorylation (secretory pathway). The nascent protein's signal peptide (1–15) routes it into the ER. In the Golgi, **Fam20C phosphorylates the Ser-x-Glu motifs** (Ser23/24/25, Ser37, cluster ~51–61), writing the acidic phosphoserine code [PMID: 23754375]. Fam20C retention and activity in the lactating gland are tuned by the Fam20A/ERGIC2/ERGIC3 system [PMID: 42439892]. The protein is then secreted into milk.

3. Micelle assembly and mineral transport (extracellular / milk). In milk, the negatively charged phosphoserine clusters of Csn1s2a bind calcium and **template/stabilize ~2 nm colloidal calcium phosphate nanoclusters** [PMID: 18420601, P 15657053], while hydrophobic C-terminal segments and chaperone-like multivalent interactions drive co-assembly with αs1-, β-, and κ-caseins into the **casein micelle** [PMID: 35815989]. κ-casein caps the surface and limits growth. The Cys96–Cys97 pair allows disulfide homo/hetero-dimerization. The result: milk holds calcium and phosphate far above solubility and delivers them to the neonate [PMID: 21245413].

4. Physiological outcome (neonate). Micellar mineral + protein nourish the pup; loss of casein causes severe neonatal growth delay [PMID: 23321409].



Classification of function. Csn1s2a is **not an enzyme** (it catalyzes no reaction), **not a membrane transporter**, and **not a signaling molecule**. It is best described as a **secreted structural/carrier phosphoprotein** whose "transport" is colloidal sequestration of calcium

phosphate, with a secondary **ATP-independent chaperone** role. Its substrate, in the loosest sense, is **calcium phosphate** (and calcium ions), coordinated through clustered phosphoserine residues.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
21245413	*The evolution of milk casein genes from tooth genes before the origin of mammals*	Establishes caseins (incl. alpha-S type) as the calcium-phosphate transport system of milk; classifies alpha-S caseins as Ca-binding/Ca-sensitive (F001)
18420601	*Supramolecular structure of the casein micelle*	Phosphoserine domains of αs2-casein stabilize CaP nanoclusters; hydrophobic domains face outward (F002, F006)
15657053	*Physicochemical characterization of CPP-ACP nanocomplexes*	Phosphoseryl clusters mediate casein–calcium-phosphate interaction (F002)
35815989	*Are casein micelles extracellular condensates formed by LLPS?*	Micelle is an extracellular assembly; caseins act as chaperones via multivalent interactions (F003, F004)
29442498	*Proteostasis... ATP-independent molecular chaperones: caseins*	Milk caseins are ATP-independent chaperones preventing aggregation/amyloid (F004)
12427545	*Structure and regulation of the murine gamma-casein gene*	Equates mouse γ-casein with ruminant αS2-casein; defines prolactin–STAT5/glucocorticoid regulation (F005, identity)
23754375	*Crystal structure of the Golgi casein kinase*	Identifies Fam20C as the SxE-motif casein kinase (F007)
24706917	*Dynamic regulation of FGF23 by Fam20C...*	Confirms Fam20C phosphorylates Ser-x-Glu motifs in secretory substrates (F007)
42439892	*Receptor-mediated Golgi retention of Fam20 kinases tunes secretome phosphorylation during lactation*	Casein phosphorylation machinery is active/required in lactation; loss reduces milk casein phosphorylation and impairs pup growth (F007)
23321409	*Behaviour of postnatally growth-impaired mice...*	α-casein-deficient milk causes severe neonatal growth delay (F008)
40845155	*Saliva protein genes shaped during primate evolution*	SCPP/casein locus diversified via duplication with the origin of lactation (F009)

Supporting context literature reviewed during the investigation (micelle structure and physical chemistry) further corroborates the nanocluster/dual-binding model, including SANS/SAXS evidence that only the nanocluster model fits micelle scattering data [PMID: 22381008], the dual-binding model of hydrophobic + colloidal-calcium-phosphate bridging [PMID: 11913691], and high-pressure/FTIR studies showing calcium phosphate release from serine-phosphate residues destabilizes the micelle [PMID: 21338142]. Mammary developmental and lactogenic regulators (Xbp1, Akt1, PI3K-AKT) were noted as upstream of the casein-synthesis program [PMID: 25713103, P 21915327], [PMID: P 32006013], consistent with Csn1s2a being a terminal lactation output.

Limitations and Knowledge Gaps

1. **No Csn1s2a-specific knockout.** The strongest loss-of-function evidence (severe neonatal growth delay) comes from **α-casein**-deficient dams [PMID: 23321409], not a targeted *Csn1s2a* knockout. The specific, non-redundant contribution of Csn1s2a — versus its paralog Csn1s2b and the other caseins — has not been isolated genetically. Functional redundancy within the micelle is likely.
2. **Phosphorylation and structural evidence are largely inferred for the mouse protein.** The four annotated phosphoserines and the Fam20C→SxE link are supported by sequence motif analysis and by experiments on β-casein/other substrates, not by direct phospho-mapping of purified mouse Csn1s2a. The second phosphocluster (~51–61) and dimerization via Cys96–Cys97 are inferences from sequence and family behavior (IBA GO terms), not direct biochemical demonstration for this protein.
3. **Chaperone/condensate role is extrapolated from the casein class.** The ATP-independent chaperone and LLPS/condensate concepts are established for caseins generally (notably αs1- and β-casein) [PMID: 29442498, P 35815989]; the quantitative chaperone efficiency of mouse Csn1s2a itself is unmeasured.

4. **Stoichiometry within the micelle is unknown.** The relative abundance of Csn1s2a vs. other caseins in mouse milk, and its precise spatial position in the micelle, have not been resolved.
5. **Minor/moonlighting functions not assessed.** Casein-derived bioactive peptides (e.g., antimicrobial, mineral-binding casein phosphopeptides) are documented for other caseins; whether mouse Csn1s2a yields physiologically relevant peptides was not experimentally examined here.

Proposed Follow-up Experiments / Actions

1. **Targeted *Csn1s2a* knockout / knock-in.** Generate a mouse line with specific *Csn1s2a* ablation (and separately a phosphoserine-to-alanine knock-in at Ser23/24/25/37) to measure the isolated contribution to milk calcium/phosphate content, micelle size/stability, and pup growth — disentangling it from Csn1s2b and α-casein.
2. **Direct phospho-mapping of purified mouse Csn1s2a.** Use LC-MS/MS on milk-purified protein to confirm the predicted phosphosites (including the ~51–61 cluster) and quantify stoichiometry, and test Fam20C dependence in *Fam20c* conditional-knockout mammary tissue.
3. **In vitro micelle reconstitution.** Reconstitute biomimetic micelles with recombinant phosphorylated vs. dephosphorylated Csn1s2a (cf. the recombinant β-casein approach in [PMID: 38318529]) to directly quantify its calcium-phosphate nanocluster stabilization and chaperone capacity.
4. **Dimerization assay.** Use non-reducing SDS-PAGE / mutagenesis of Cys96/Cys97 to confirm disulfide-linked homodimer/heterodimer formation and assess its role in micelle architecture.
5. **Promoter/ChIP validation.** ChIP-seq for STAT5 and glucocorticoid receptor at the *Csn1s2a* locus in lactating mammary tissue to confirm direct occupancy of the predicted sites [PMID: 12427545].
6. **Structural prediction with confidence analysis.** Run AlphaFold on Q02862 and interpret pLDDT/PAE to formally confirm intrinsic disorder of the phosphoserine/acidic regions and characterize any ordered segments around the cysteine pair.

Conclusion

Mouse **Csn1s2a (Q02862)** is a secreted, mammary-gland-specific milk phosphoprotein of the calcium-sensitive alpha-S2 casein family. Its primary function is **structural/nutritional, not catalytic**: clustered N-terminal Ser-x-Glu phosphoserines (written by the Golgi casein kinase Fam20C) bind and sequester calcium phosphate, allowing co-assembly into the **extracellular casein micelle** and enabling milk to transport calcium and phosphate to the neonate for skeletal and dental mineralization. It acts **outside the cell** in secreted milk, its transcription is driven by the **prolactin–STAT5 / glucocorticoid** lactation program, and its intrinsic disorder additionally provides **ATP-independent chaperone** activity. Genetic loss of casein from mouse milk causes severe neonatal growth delay, confirming the family's essential nutritional-delivery role.